

LBA-Net: Boundary-Aware Self-Distillation for Efficient Semantic Segmentation

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Abstract

Lightweight semantic segmentation models are essential for deployment in resource-constrained environments, yet limited representational capacity often leads to degraded boundary localization and structural inconsistencies. Existing approaches commonly address this by increasing architectural complexity, sacrificing efficiency or generality. We propose LBA-Net (Lightweight Boundary-Attentive Network), a lightweight and general-purpose segmentation framework that combines boundary-aware decoding with self-distillation via an exponential moving average (EMA) teacher. LBA-Net introduces a boundary-attentive decoder that explicitly disentangles semantic context and boundary refinement through parallel decoding pathways, injecting structural inductive bias into the reconstruction process. To stabilize optimization in low-capacity encoders, EMA-based self-distillation is employed without requiring a high-capacity external teacher or increasing inference cost. Experiments across natural and medical image segmentation benchmarks demonstrate that LBA-Net achieves strong accuracy and efficiency trade-offs, supporting its effectiveness for efficient dense prediction.

1. Introduction

Semantic segmentation is a fundamental problem in computer vision, requiring dense pixel-level predictions that accurately delineate object regions and boundaries. It underpins a wide range of applications, including autonomous driving, scene understanding, robotics, and medical image analysis. Despite significant progress driven by deep convolutional neural networks (CNNs) and attention-enhanced architectures (Woo et al., 2018), segmentation remains challenging due to large intra-class variability, fine-grained struc-

tural details, and the need to balance accuracy with computational efficiency.

Encoder-decoder architectures dominate modern segmentation pipelines, with U-Net (Ronneberger et al., 2015) and its variants forming a common foundation. These methods combine hierarchical feature extraction with progressive spatial reconstruction via skip connections. However, when lightweight backbones are employed to reduce model complexity, standard encoder-decoder designs often struggle to recover precise object boundaries. This issue is particularly pronounced for thin structures, small objects, and low-contrast regions, where aggressive downsampling leads to over-smoothed predictions.

To mitigate these limitations, recent works incorporate attention mechanisms, multi-scale context aggregation, and transformer-based global modeling (Cao et al., 2022; Valanarasu et al., 2021; Roy et al., 2023). While effective, such approaches introduce substantial computational and memory overhead, restricting deployment in resource-constrained environments. This challenge is especially relevant in medical imaging, where imaging workloads continue to grow while expert availability and computational resources remain limited (The World Bank, 2022; Royal College of Radiologists, 2023).

Another complementary research direction focuses on boundary-aware segmentation, motivated by the observation that segmentation errors are frequently concentrated near object contours. Prior studies demonstrate that explicit boundary modeling can improve structural accuracy in both medical and natural image segmentation tasks (Candemir et al., 2014; Bernal et al., 2012; Lou et al., 2023; Liu et al., 2024). However, in many existing methods, boundary information is introduced only as auxiliary supervision rather than being tightly integrated into the feature reconstruction process, limiting its effectiveness in lightweight architectures.

In parallel, knowledge distillation has emerged as an effective strategy for improving the performance and generalization of compact models. Classical distillation relies on high-capacity teachers, while more recent self-distillation approaches based on exponential moving average (EMA) teachers provide temporally smoothed supervision without external models (Tarvainen & Valpola, 2017; Bai &

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Xia, 2022). Although successful in classification and semi-supervised learning, the integration of EMA-based self-distillation with boundary-aware segmentation architectures remains underexplored.

In this work, we argue that the limitations of lightweight segmentation models stem from two interrelated factors: insufficient structural inductive bias for boundary reasoning and unstable optimization dynamics in low-capacity networks. To address these challenges, we propose **LBA-Net**, a lightweight boundary-attentive segmentation framework regularized via EMA-based self-distillation. LBA-Net explicitly disentangles semantic and boundary information through parallel decoding pathways while enforcing prediction consistency between student and teacher networks. Unlike transformer-heavy models or conventional distillation approaches, LBA-Net maintains a compact architecture and does not require a high-capacity external teacher.

We evaluate LBA-Net on diverse natural and medical image segmentation benchmarks, including Pascal VOC, ADE20K, and multiple medical datasets. Extensive experiments demonstrate that LBA-Net achieves favorable accuracy-efficiency trade-offs compared to both lightweight and heavyweight baselines, highlighting its effectiveness as a general-purpose solution for efficient and boundary-sensitive dense prediction.

The main contributions of this work are summarized as follows:

- We propose a lightweight encoder-decoder segmentation framework with a boundary-attentive decoder that explicitly preserves fine-grained contour information.
- We integrate an EMA-based self-distillation strategy to improve training stability and generalization without relying on an external teacher.
- We conduct extensive evaluations on diverse natural and medical image datasets, demonstrating consistent accuracy and efficiency improvements.
- We provide comprehensive ablation studies to analyze the impact of architectural components, backbone choices, and boundary modeling.

2. Related Work

2.1. Encoder-Decoder Architectures for Semantic Segmentation

Encoder-decoder architectures form the foundation of modern semantic segmentation. U-Net (Ronneberger et al., 2015) introduced skip connections to combine low-level spatial details with high-level semantic representations, enabling accurate localization while preserving contextual in-

formation. This paradigm has inspired numerous extensions incorporating residual connections, dense feature reuse, and multi-scale aggregation to improve representational capacity and optimization stability (Diakogiannis et al., 2020; Shah et al., 2023). Despite their success, repeated downsampling and upsampling in standard encoder-decoder designs often lead to boundary blurring, particularly when lightweight encoders are employed.

2.2. Attention and Transformer-Based Segmentation

Attention mechanisms have been widely explored to improve segmentation by selectively emphasizing informative spatial and channel-wise features. Early approaches demonstrated that spatial gating and feature reweighting can enhance localization, while later methods introduced axial attention, cross-feature fusion, and adaptive aggregation strategies (Lou et al., 2023; Jian et al., 2025). More recently, transformer-based architectures model long-range dependencies through self-attention and achieve strong segmentation performance (Valanarasu et al., 2021; Cao et al., 2022; Roy et al., 2023). However, the computational and memory overhead of transformers often limits their suitability for resource-constrained or real-time deployment.

2.3. Boundary-Aware and Structure-Guided Segmentation

Accurate boundary delineation remains a key challenge in segmentation, as most errors occur near object contours. Early works incorporated anatomical priors and atlas-based modeling (Candemir et al., 2014), while subsequent deep learning approaches introduced explicit boundary supervision and edge-aware learning to improve structural accuracy (Bernal et al., 2012; 2015; Fernández-Esparrach et al., 2016). Recent methods such as CaraNet (Lou et al., 2023), IECFNet (Liu et al., 2024), TCFMA-Net (Manan et al., 2025), and ARAA-Net (Deng et al., 2024) integrate attention and multi-scale fusion to enhance boundary precision in challenging scenarios involving small or overlapping structures. Nevertheless, many of these approaches rely on complex multi-branch designs or treat boundary learning as an auxiliary objective, increasing computational cost and limiting scalability.

2.4. Lightweight Segmentation Models

The demand for efficient segmentation has driven the development of lightweight architectures based on depthwise separable convolutions and compact encoders. MobileNet-based backbones offer favorable accuracy-efficiency trade-offs for image-level recognition but often underperform in dense prediction tasks, producing fragmented regions and blurred boundaries. Several studies address this limitation through channel recalibration, adaptive fusion, and attention-

based refinement; however, achieving accurate boundary localization and structural consistency in low-capacity models remains an open challenge.

2.5. Knowledge Distillation and Self-Distillation

Knowledge distillation improves generalization by transferring knowledge from a teacher model to a student model. Traditional frameworks rely on high-capacity teachers, which are often impractical for lightweight or deployment-oriented settings. Self-distillation approaches based on exponential moving average (EMA) teachers, such as the Mean Teacher framework (Tarvainen & Valpola, 2017), eliminate the need for external teachers by providing temporally smoothed supervision. While EMA-based self-distillation has shown strong performance in classification and semi-supervised learning, its application to dense prediction tasks, particularly in combination with boundary-aware architectures, remains underexplored.

2.6. Positioning of This Work

LBA-Net addresses boundary precision, computational efficiency, and training stability within a unified framework. Unlike transformer-heavy models, it maintains a compact architecture suitable for resource-constrained deployment. Unlike conventional boundary-aware approaches, it embeds boundary reasoning directly into the decoding process via a dual-branch design. Finally, unlike classical distillation frameworks, it employs EMA-based self-distillation without requiring an external high-capacity teacher. This combination enables favorable accuracy-efficiency trade-offs and robust generalization across diverse segmentation domains.

3. Proposed Method

3.1. Architecture Overview

LBA-Net (Lightweight Boundary-Attentive Network) is a compact encoder-decoder segmentation framework designed to preserve structural accuracy while maintaining computational efficiency.

Let $\mathcal{D} = \{(x_i, y_i, b_i)\}_{i=1}^N$ denote a dataset, where $x_i \in \mathbb{R}^{H \times W \times C}$ is an input image, $y_i \in \{1, \dots, K\}^{H \times W}$ is the corresponding ground-truth segmentation mask, $b_i \in \{0, 1\}^{H \times W}$ is a binary boundary map derived from y_i using morphological gradient operations, and N denotes the number of training samples.

The segmentation model is defined as a lightweight mapping function

$$f_\theta : x \mapsto (p, q),$$

where $p \in [0, 1]^{H \times W \times K}$ denotes per-pixel class probabilities, θ denotes the learnable network parameters, and $q \in [0, 1]^{H \times W}$ denotes boundary likelihoods, under strict

computational and memory constraints.

The architecture integrates three core components: (i) an efficient encoder with channel recalibration, (ii) a boundary-attentive decoder that explicitly disentangles semantic and boundary information, and (iii) an exponential moving average (EMA) based self-distillation mechanism that stabilizes training without requiring an external teacher.

An overview of the proposed architecture is shown in Figure 1. As illustrated in Figure 1, boundary prediction and semantic segmentation are learned jointly within a unified decoder, while training stability is enforced through EMA-based self-distillation. The student network is optimized using supervised and consistency losses, while the teacher network shares the same architecture and is updated solely via EMA parameter averaging.

3.2. EMA-Based Self-Distillation

To improve training stability and representation consistency in low-capacity settings, LBA-Net adopts an EMA-based self-distillation strategy. Let θ_s and θ_t denote the parameters of the student and teacher networks, respectively. The teacher parameters are updated at iteration k as

$$\theta_t^{(k)} = \alpha \theta_t^{(k-1)} + (1 - \alpha) \theta_s^{(k)}, \quad (1)$$

where $\alpha \in [0, 1)$ controls the temporal smoothing factor.

Both the student and teacher networks produce per-pixel logits for segmentation and boundary prediction, which are normalized using a channel-wise softmax operation. During training, the student’s soft probability outputs are aligned with the teacher’s temporally smoothed soft labels via a distillation loss, while hard predictions are supervised using ground-truth masks. The teacher network is updated exclusively through exponential moving average (EMA) and does not receive gradient updates, introducing no additional inference-time overhead while improving optimization stability and generalization.

3.3. Encoder with Channel Recalibration

The encoder extracts hierarchical feature representations at progressively coarser spatial resolutions. We adopt MobileNetV2 as the backbone due to its favorable accuracy and efficiency trade-off. However, depthwise separable convolutions exhibit limited inter-channel interaction. To compensate for this limitation, each encoder stage is augmented with a Squeeze-and-Excitation (SE) block as shown in Figure 2.

Given an input feature map $X \in \mathbb{R}^{C \times H \times W}$, the SE block performs:

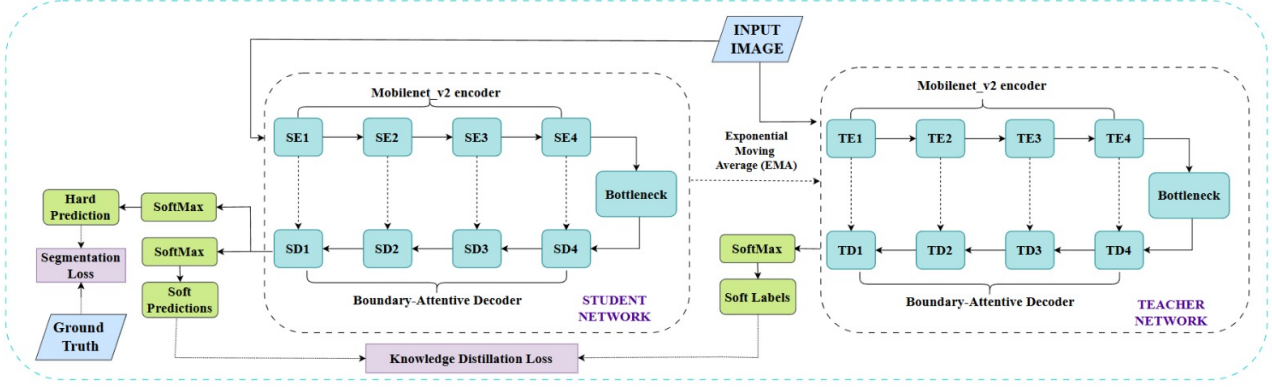


Figure 1. Overview of the proposed LBA-Net architecture. The framework follows a student-teacher paradigm, where the teacher network shares the same architecture as the student and is updated via exponential moving average (EMA) of the student parameters. The student is trained using ground-truth supervision and a consistency loss with the teacher’s predictions.

Squeeze:

$$z_c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W X_c(i, j), \quad (2)$$

Excitation:

$$s = \sigma(W_2 \text{ReLU}(W_1 z)), \quad (3)$$

where $z = [z_1, \dots, z_C]^T \in \mathbb{R}^C$ denotes the channel-wise descriptor obtained after global average pooling, $W_1 \in \mathbb{R}^{\frac{C}{r} \times C}$, $W_2 \in \mathbb{R}^{C \times \frac{C}{r}}$, r is the reduction ratio, and $\sigma(\cdot)$ denotes the sigmoid activation.

Recalibration:

$$\tilde{X}_c = s_c \cdot X_c. \quad (4)$$

where s_c denotes the learned channel-wise excitation weight and X_c denotes the c -th channel of the input feature map.

This operation enhances informative channels while suppressing redundant responses with negligible computational overhead.

3.4. Bottleneck Representation

The bottleneck aggregates high-level semantic information before decoding:

$$b = \text{Bottleneck}(e_4), \quad (5)$$

where e_4 denotes the deepest encoder feature map. Convolution, batch normalization, and ReLU activation jointly produce a compact and discriminative representation for subsequent spatial reconstruction. The bottleneck operates at the lowest spatial resolution and serves as a compact semantic representation that bridges the encoder and decoder.

3.5. Boundary-Attentive Decoder

The decoder progressively reconstructs a high-resolution segmentation map using a sequence of Boundary Decoder Blocks (BDBs). At decoding stage i , features are formed as:

$$d_i^{\text{in}} = \text{Concat}[\text{Upsample}(d_{i+1}), e_i]. \quad (6)$$

where e_i denotes the encoder feature map at stage i , and d_{i+1} denotes the decoder output from the previous coarser scale.

Each BDB employs a dual-branch design consisting of (i) a **semantic branch** for contextual feature extraction and (ii) a **boundary branch** for preserving high-frequency spatial details.

Let the fused feature map be denoted as F . Channel-wise pooled representations are first computed as

$$F_{\text{avg}} = \text{AvgPool}_{\text{ch}}(F), \quad F_{\text{max}} = \text{MaxPool}_{\text{ch}}(F), \quad (7)$$

where $\text{AvgPool}_{\text{ch}}(\cdot)$ and $\text{MaxPool}_{\text{ch}}(\cdot)$ denote channel-wise average and max pooling operations, respectively.

The spatial attention map is then obtained as

$$M_{\text{spatial}} = \sigma(\text{Conv}_{5 \times 5}(\text{Concat}[F_{\text{avg}}, F_{\text{max}}])). \quad (8)$$

The refined feature representation is obtained via element-wise multiplication with M_{spatial} , enabling the decoder to emphasize boundary-relevant regions during spatial reconstruction.

3.6. Learning Objective

The overall training objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \lambda_b \mathcal{L}_{\text{boundary}} + \lambda_{kd} \mathcal{L}_{\text{kd}}. \quad (9)$$

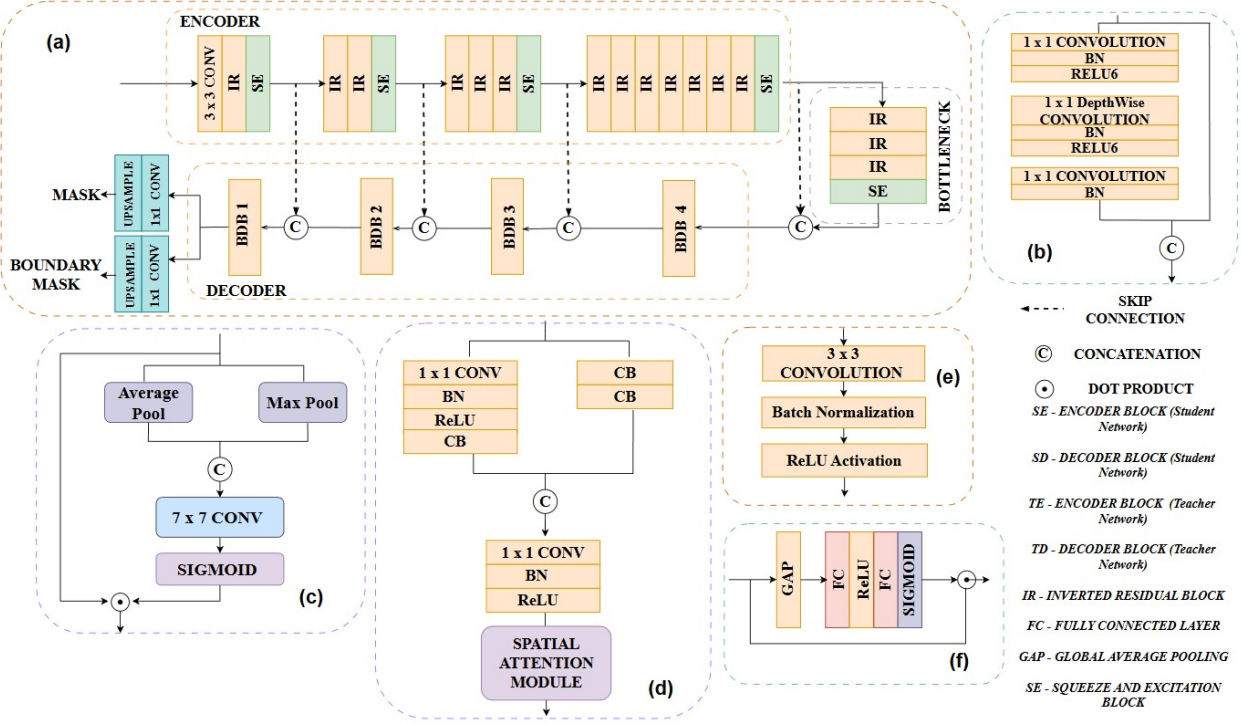


Figure 2. Structural components of the proposed LBA-Net encoder-decoder framework. (a) Overall hierarchical encoder-decoder architecture employing Inverted Residual (IR) blocks and Boundary Decoder Blocks (BDB), producing segmentation and boundary predictions at the decoder output. (b) Inverted Residual (IR) block with depthwise separable convolutions and residual connections. (c) Spatial attention module using parallel average and max pooling. (d) Boundary Decoder Block (BDB) integrating multi-scale features with spatial attention. (e) Standard convolution block (CB) consisting of convolution, batch normalization, and ReLU. (f) Squeeze-and-Excitation (SE) block for channel-wise feature recalibration.

Segmentation Loss:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{focal}} + \mathcal{L}_{\text{dice}}.$$

Boundary Loss: We define the per-pixel boundary loss as

$$\ell_i^b = -[y_i^b \log \sigma(p_i^b) + (1 - y_i^b) \log (1 - \sigma(p_i^b))], \quad (10)$$

where y_i^b and p_i^b denote the ground-truth and predicted boundary logits at pixel i , respectively.

The overall boundary loss is then computed as

$$\mathcal{L}_{\text{boundary}} = \frac{1}{|B|} \sum_{i \in B} \ell_i^b, \quad (11)$$

where B denotes the set of boundary pixels.

Distillation Loss:

$$\mathcal{L}_{\text{kd}} = \frac{1}{N} \sum_{i=1}^N \|p_s^{(i)} - p_t^{(i)}\|_2^2. \quad (12)$$

where p_s and p_t denote the softmax-normalized prediction maps of the student and teacher networks, respectively.

This consistency constraint aligns student predictions with temporally smoothed teacher outputs, improving robustness and generalization in lightweight segmentation models.

4. Experiments and Results

Our experimental analysis evaluates LBA-Net from three perspectives: (i) comparison with state-of-the-art lightweight and heavyweight models, (ii) contribution of individual architectural components via ablation studies, and (iii) generalization across datasets with diverse image statistics, object scales, and label structures.

4.1. Datasets

Medical Image Segmentation: We evaluate LBA-Net on three representative medical image segmentation benchmarks that emphasize boundary precision, low-contrast structures, and small-object delineation:

- **THRS-RSNA:** The THRS-RSNA dataset is derived from the RSNA Pediatric Bone Age Challenge Dataset and consists of two subsets—epiphyseal and articular surface. It contains 1068 hand radiographs spanning

ages 9 to 228 months, with expert-annotated segmentation masks for epiphyseal and articular surface regions (Deng et al., 2024).

- **Montgomery County:** Developed by the National Library of Medicine (NLM), the Montgomery County dataset (Candemir et al., 2014; Jaeger et al., 2014) contains 138 posterior–anterior chest X-ray images manually segmented and annotated for lung fields and tuberculosis manifestations. It is a recognized benchmark for lung boundary extraction and TB screening, and is widely used in pulmonary disease imaging research (Zhang et al., 2024; Shah et al., 2023).
- **Kvasir-SEG:** Kvasir-SEG (Jha et al., 2019) is an open-access dataset of 1000 gastrointestinal polyp images with corresponding pixel-wise segmentation masks manually annotated and verified by an experienced gastroenterologist. Images vary in resolution (332×487 to 1920×1072 pixels) and are provided with binary masks and bounding box annotations. The dataset is designed to advance polyp segmentation and reduce miss rates in colonoscopy (14%–30%), supporting research in detection, localization, and classification (Wang et al., 2024; Jian et al., 2025; Manan et al., 2025).

Natural Image Segmentation: To evaluate domain-agnostic generalization, we assess LBA-Net on three standard natural image segmentation benchmarks:

- **Pascal VOC 2012** consists of 21 semantic classes (20 foreground and one background), with label value 255 treated as an ignore index. We follow the official dataset splits with 1,464 training and 1,449 validation images. The model is trained on the training split and evaluated on the combined trainval set of 2,913 images, following prior lightweight segmentation benchmarks. A reduced background class weight of 0.1 is used to mitigate class imbalance. All images are resized to 512×512 .
- **ADE20K** is a large-scale scene parsing dataset with dense annotations. We evaluate on a remapped 5-class subset, using 17,768 training, 4,442 validation, and 2,000 test images, resized to 428×428 .
- **Cityscapes** focuses on urban scene understanding using street-level imagery. We adopt a reduced 5-class subset following standard evaluation protocols, with an ignore index of 255. The dataset includes 2,980 training, 500 validation, and 1,525 test images at a resolution of 1024×2048 .

Evaluation on ADE20K and Cityscapes provides a stringent test for lightweight segmentation models and verifies that the

proposed boundary-aware decoding and EMA-based self-distillation generalize beyond medical imaging domains.

4.2. Evaluation Metrics

We report standard segmentation metrics: mean Intersection-over-Union (mIoU), mean Dice coefficient (mDice), Overall Accuracy (OA), and Frequency-Weighted IoU (FWIoU). For efficiency analysis, we additionally report the number of parameters (Params), floating point operations (FLOPs), and inference speed (FPS).

4.3. Implementation Details

All models are implemented using PyTorch and trained on NVIDIA GPUs. Images are resized to dataset-specific fixed resolutions while preserving aspect ratios. We use the Adam optimizer with an initial learning rate of 1×10^{-4} , batch size of 10, and train for 500 epochs. The EMA decay factor for self-distillation is set to $\alpha = 0.999$. Loss weights are fixed to $\lambda_b = 2.0$ for boundary supervision and $\lambda_{kd} = 1.0$ for distillation. Inference speed (FPS) is measured on a single NVIDIA GPU with batch size 1.

Data augmentation includes random flipping, rotation, elastic deformation, intensity perturbation, and affine transformations for medical datasets. The same architecture and training protocol are used across all datasets without domain-specific tuning.

4.4. Comparison with Existing Models

Table 1 reports quantitative comparisons on the THRS-RSNA epiphysis segmentation tasks. Additional quantitative results for articular surface segmentation on THRS-RSNA are provided in the supplementary material. LBA-Net achieves the highest mIoU and mDice among all compared methods while using significantly fewer parameters and FLOPs than transformer-based and multi-attention architectures.

Notably, LBA-Net outperforms ARAA-Net while reducing the parameter count by more than an order of magnitude, demonstrating that explicit boundary modeling and self-distillation can effectively compensate for reduced model capacity.

4.5. Multi-Dataset Evaluation

Table 2 and Table 3 present a comprehensive comparison on colonoscopy and chest X-ray datasets. LBA-Net consistently outperforms classical U-Net variants and recent boundary-aware models in both Dice and IoU metrics, particularly on datasets with small targets and ambiguous contours.

These results indicate that the boundary-attentive decoder

Table 1. Quantitative results for epiphysis segmentation on the THRS-RSNA dataset, including accuracy and efficiency metrics.

Method	mIoU (%)	mDice (%)	FWIoU (%)	OA (%)	FPS	FLOPs (G)	Params (M)
CaraNet	85.01	85.67	98.38	99.07	16.49	29.12	44.590
GAA	86.67	83.19	98.47	99.22	4.04	5.93	1.370
HyperSeg	87.76	85.75	98.57	99.25	2.89	1.55	3.440
JCS	88.71	86.75	98.62	99.25	95.51	243.02	42.500
Swin-Unet	90.07	88.39	98.83	99.39	47.40	78.98	27.150
ARAA-Net	91.65	89.91	99.16	99.56	2.33	302.25	414.630
LBA-Net (Ours)	91.86	90.51	99.24	99.60	55.21	22.872	13.880

Table 2. Evaluation on medical benchmarks. mD: mean Dice, mI: mean IoU.

Data	Model	mD	mI
Mont.	Lung Atlas (Candemir et al., 2014)	0.960	-
	ARAA-Net (Deng et al., 2024)	0.967	0.959
	LBA-Net	0.988	0.986
Kvasir	U-Net (Ronneberger et al., 2015)	0.818	0.746
	IECFNet (Liu et al., 2024)	0.907	0.856
	CaraNet (Lou et al., 2023)	0.918	0.865
	TCFMA-Net (Manan et al., 2025)	0.915	0.870
	BiCoMA (Khan et al., 2025)	0.929	0.883
	HarDNet-CPS (Wang et al., 2024)	0.911	0.856
	Siam. Dif. (Qiu et al., 2025)	0.912	0.855
	LBA-Net	0.950	0.962

effectively preserves fine-scale structures even under challenging imaging conditions.

4.6. Ablation Studies

Effect of Encoder Backbone: Table 4 evaluates different encoder backbones within the LBA-Net framework. MobileNetV2 achieves the best accuracy and efficiency trade-off, outperforming heavier backbones such as ResNet-50 while using fewer parameters.

Effect of Architectural Components: Table 5 analyzes the contribution of the Boundary Decoder Block (BDB). Replacing standard convolutions with depthwise variants reduces FLOPs by approximately 29% but results in a small accuracy drop, confirming that the proposed BDB design is critical for precise boundary reconstruction.

4.7. Qualitative Results

Figure 3 presents qualitative segmentation examples. Compared to competing methods, LBA-Net produces sharper contours, fewer boundary artifacts, and more structurally consistent masks, particularly for thin and low-contrast regions. These visual results corroborate the quantitative improvements reported above.

Table 3. Evaluation on non-medical benchmarks. Results on ADE20K are reported for the selected 5-class subset. mI: mean IoU.

Model	mI
<i>PASCAL VOC 2012</i>	
HRNet (Sun et al., 2019)	0.800
DeepLabV3+ (Chen et al., 2018)	0.890
PSPNet (Zhao et al., 2017)	0.854
Pseudo-SD (Zhao et al., 2025)	0.827
Fill-UNet (Liu et al., 2025)	0.833
SegNeXt-T (Guo et al., 2022)	0.827
LBA-Net	0.880
<i>ADE20K (5-class subset)</i>	
GMMSeg (Liang et al., 2022)	0.490
ProtoSeg (Zhou et al., 2022)	0.486
UperNet (Xiao et al., 2018)	0.489
SegFormer (Xie et al., 2021)	0.489
LBA-Net	0.914
<i>Cityscapes</i>	
SBCB (DDS) (Ishikawa & Aoki, 2023)	0.806
SBCB (CASENet) (Ishikawa & Aoki, 2023)	0.820
PSPNet (Zhao et al., 2017)	0.802
DeepLabV3+ (Chen et al., 2018)	0.821
Pseudo-SD (Zhao et al., 2025)	0.815
LBA-Net	0.806

4.8. Cross-Domain Generalization

Table 6 summarizes five-fold cross-validation results across medical and natural image datasets. LBA-Net maintains consistently strong performance across low-contrast chest X-rays, boundary-sensitive colonoscopy images, anatomical hand structures, and complex multi-class natural scenes. The consistent high scores across mIoU, mDice, FWIoU, and OA reinforce the model’s robustness.

Importantly, no architectural or hyperparameter changes are introduced across domains, except for adjusting the input image size for non-medical datasets, highlighting the robustness and generality of the proposed framework.

Table 4. Ablation study on different encoder backbones for epiphysis segmentation on the THRS-RSNA dataset

Backbone	mIoU(%)	OA(%)	mDice(%)
MobileNetV2 (Ours)	91.84	99.60	90.55
ResNet-50	89.41	99.33	87.24
VGG-19	78.06	99.49	70.84
VGG-16	91.57	99.58	90.27

Table 5. Ablation study on architectural components for epiphysis segmentation on the THRS-RSNA dataset. mI and mD denote mIoU and mDice, respectively. F (G) and Par (M) indicate computational cost in GFLOPs and model size in millions of parameters. DW-BDB refers to the Depthwise Boundary Decoder Block.

Method	mI (%)	mD (%)	F (G)	Par (M)
LBA-Net	91.84	90.55	22.87	13.88
LBA-Net + DW-BDB	91.16	89.70	16.19	12.95
LBA-Net + DW-BDB + Tversky	90.57	88.96	16.19	12.95

Table 6. Five-fold cross-validation results of the proposed LBA-Net across multiple datasets.

Dataset	mIoU	mDice	FWIoU	OA
Montgomery (Candemir et al., 2014)	0.9807	0.9837	0.9864	0.9931
Kvasir (Jha et al., 2019)	0.9386	0.9359	0.9689	0.9839
Epiphysis (Deng et al., 2024)	0.9184	0.9062	0.9924	0.9960
Articular (Deng et al., 2024)	0.9015	0.9012	0.9803	0.9897
Pascal VOC 2012 (Everingham et al., 2012)	0.7361	0.9763	0.9078	0.9471
Cityscapes (Cordts et al., 2016)	0.6932	0.9999	0.9502	0.9727
ADE20K (Zhou et al., 2017; 2019)	0.9305	0.8890	0.9667	0.9827

5. Analysis and Discussion

LBA-Net improves lightweight segmentation by explicitly addressing boundary localization and training stability. The dual-branch Boundary Decoder Block (BDB) separates semantic context modeling from boundary refinement, preserving high-frequency structural details typically lost due to aggressive downsampling, with empirical gains concentrated near object boundaries.

LBA-Net achieves a favorable accuracy-efficiency trade-off, delivering competitive or superior performance with substantially fewer parameters and lower computational cost than transformer-based alternatives. EMA-based self-distillation further stabilizes optimization and improves representation consistency in low-capacity encoders without increasing inference complexity or requiring an external teacher. Together, these design choices enable robust cross-domain

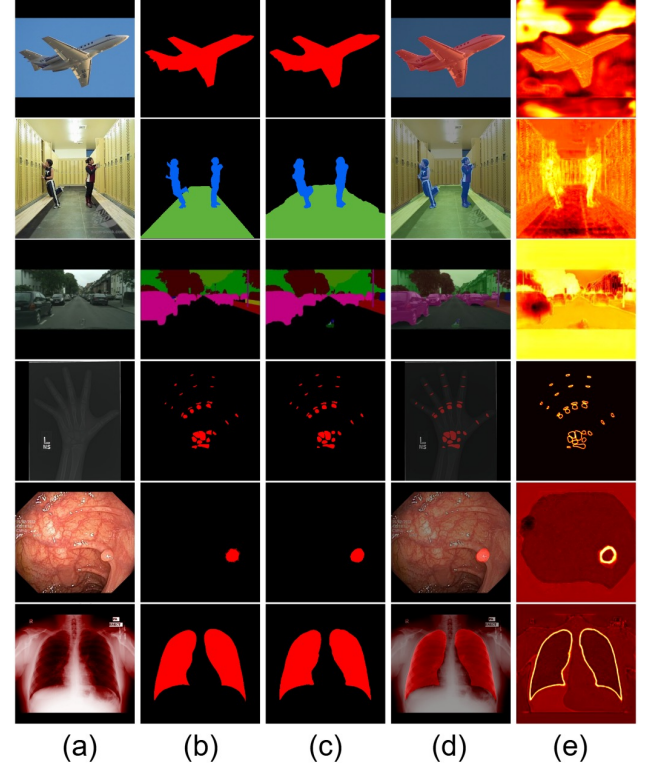


Figure 3. Qualitative segmentation results on multiple datasets from top to bottom: Pascal VOC 2012, ADE20K, Cityscapes, THRS-RSNA, Kvasir-SEG, and Montgomery. From left to right: (a) input image, (b) ground-truth mask, (c) predicted segmentation, (d) prediction overlay, and (e) predicted boundary map.

generalization across medical and natural image datasets without domain-specific architectural modifications.

6. Conclusion

This paper introduced **LBA-Net**, a lightweight and boundary-aware segmentation framework that balances accuracy, computational efficiency, and generalization. LBA-Net integrates three components: a compact MobileNetV2-based encoder with channel recalibration, a dual-branch boundary-aware decoder that preserves fine-grained structural information, and an EMA-based self-distillation strategy that stabilizes training without increasing inference cost.

Extensive experiments across diverse segmentation benchmarks show that LBA-Net achieves competitive or superior accuracy compared to both lightweight and heavyweight baselines while requiring substantially fewer parameters and lower computational cost. Explicit boundary modeling combined with consistency-based regularization preserves structural accuracy in challenging thin, low-contrast, and overlapping regions, while enabling robust generalization across heterogeneous datasets without domain-specific architectural modifications.

Impact Statement

This work studies architectural and training strategies for improving the accuracy and efficiency trade-off in semantic segmentation. The proposed LBA-Net is intended as a general-purpose segmentation framework and does not constitute a clinical decision-making system.

By reducing computational requirements while preserving structural accuracy, the method may facilitate deployment in resource-constrained settings and support downstream computer vision applications. The approach relies solely on publicly available datasets and does not introduce new data collection or privacy risks.

As with any data-driven model, performance may vary across domains and datasets, and appropriate validation is required before use in high-stakes applications.

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